

MH371 truth-blind cruise control

mh370_analog_prior_observed_bfo | 2 deterministic seeds | generated from run artifacts

Control result

MODEL FAMILY	PARTICLES
mh370_analog_prior_observed_bfo	12000 x 2 seeds
TERMINAL MEAN ERROR	TERMINAL MASS WITHIN 100 NM
28.4 to 45.6 NM	73.4 to 85.1%
TERMINAL ROOT ESS	BFO OBSERVATION SD
16.3 to 20.9	1.5 Hz
TRUTH-IMPLIED BFO BIAS	BTO EQUATION CLOSURE
167.2 to 171.6 Hz	3.076e-1 us max

Evidence and inference conditions

- Cruise only: 01:48 UTC initial state through the selected 06:48:33 UTC SATCOM observation.
- Later ACARS positions, trajectory, Mach, altitude, and headings were excluded from inference and loaded only here.
- ERA5 temperature and wind plus IGRF-14 east-positive magnetic declination drive every propagation step.
- Satellite oscillator and Perth GES AFC terms remain separate inputs; aircraft BFO bias is analytically estimated.
- Each map combines deterministic seed replicates with equal replicate weight.

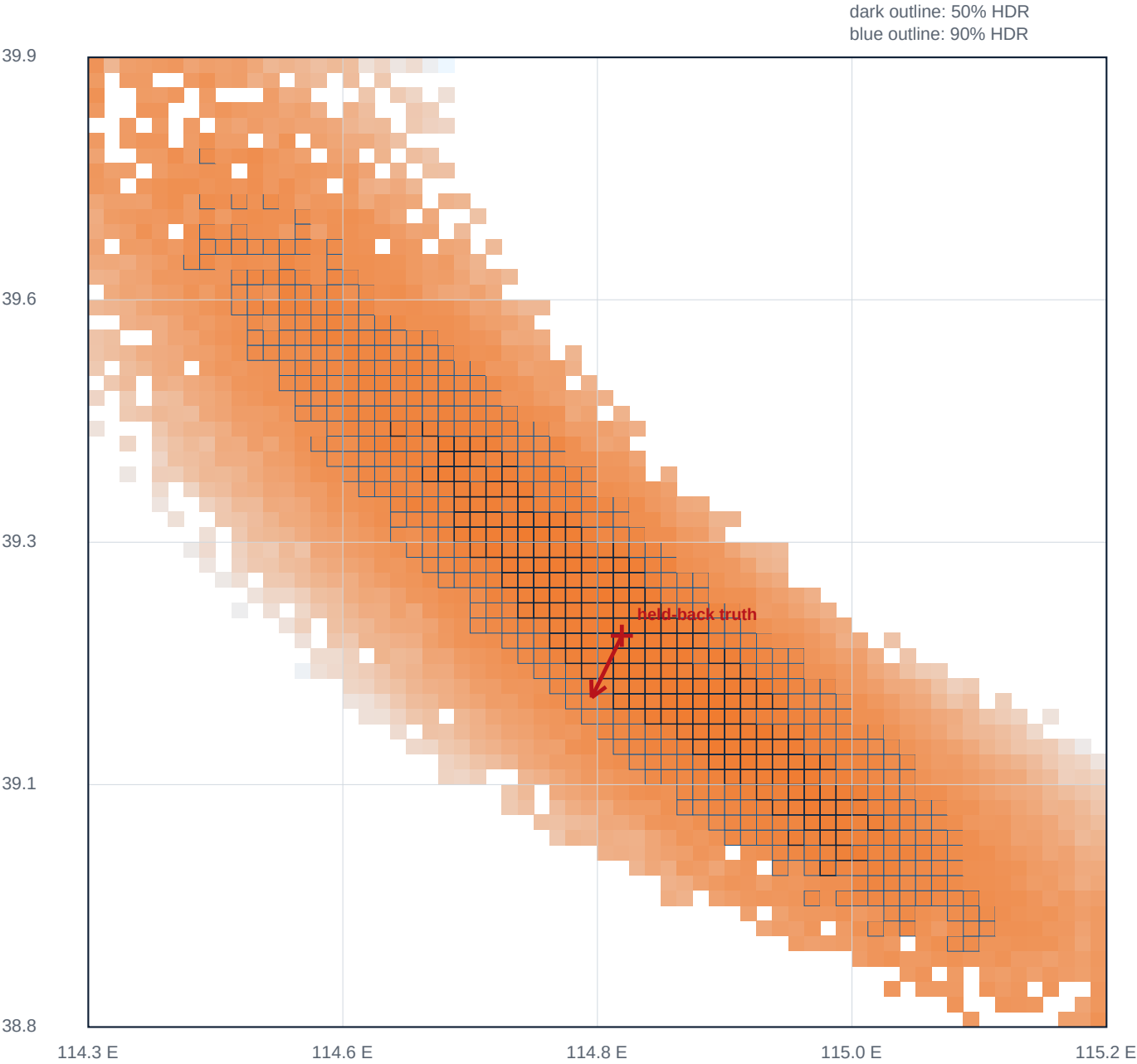
Material limitations

- held-back ACARS positions and trajectories were loaded only by this scorer
- posterior localization is conditional on the declared motion and BFO-noise family
- this known-flight control does not establish adequacy for the MH370 accident flight

The following 6 pages show posterior density, held-back position, and true-heading arrow at each selected SATCOM arc.

mh371_0200 SATCOM arc

2014-03-07T01:56:47.921Z | truth 39.2204 deg, 114.7725 deg | true heading 206.0 deg



Posterior and control metrics

MEAN POSITION ERROR
2.0 NM

MACH: POSTERIOR / TRUTH
0.783 / 0.759

GROUND SPEED: POSTERIOR / TRUTH
388 / 383 kt

TRUTH-IMPLIED BFO BIAS
167.2 Hz

POSTERIOR WITHIN 100 NM
100.0%

ALTITUDE: POSTERIOR / TRUTH
34263 / 27599 ft

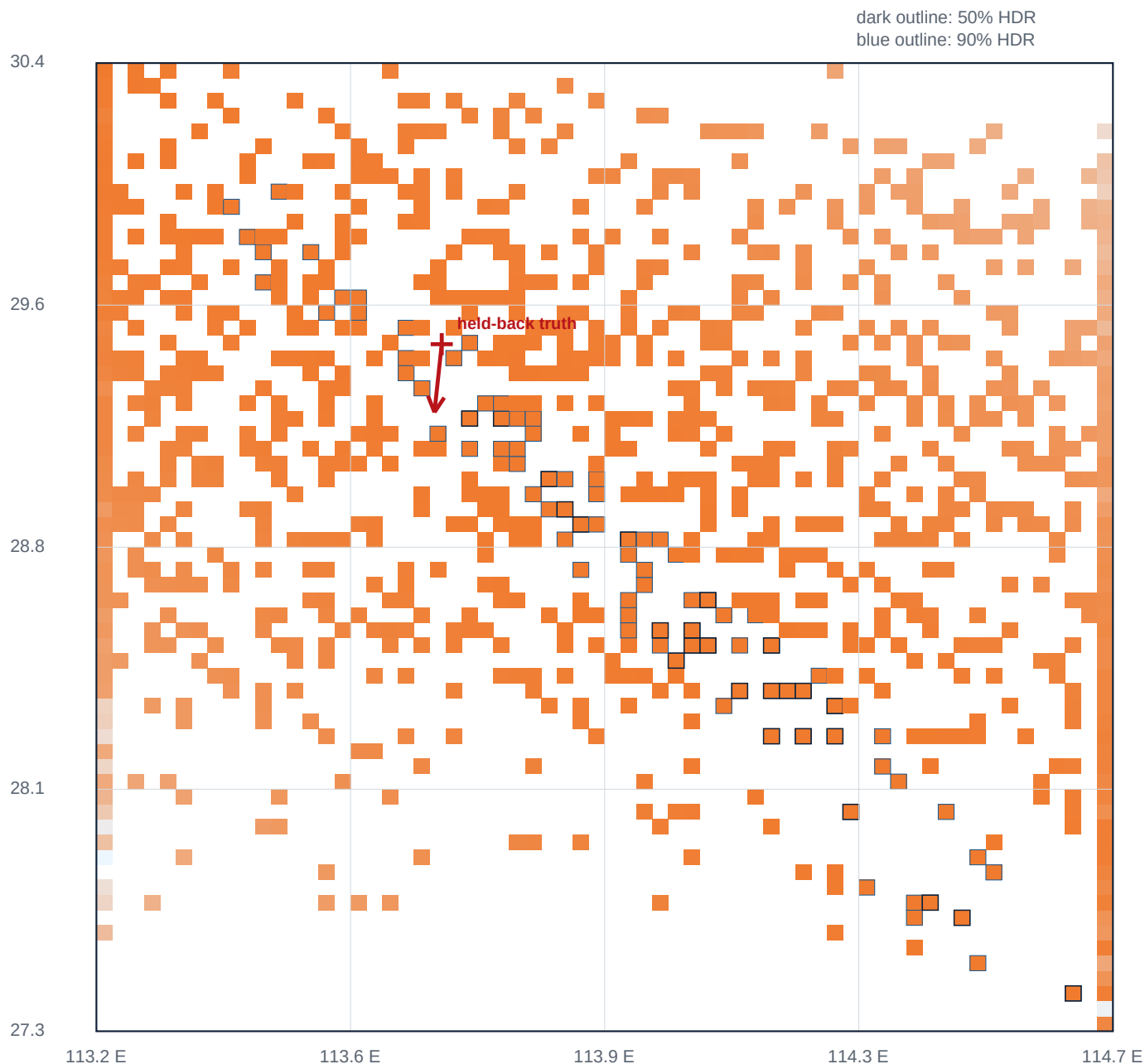
LATENT BFO BIAS
167.3 Hz

PARTICLE / ROOT ESS
7782 / 7782.4

Density combines equal-weight deterministic seed replicates. The red cross and arrow were loaded only during scoring.

mh371_0300 SATCOM arc

2014-03-07T03:20:22.428Z | truth 29.4730 deg, 113.6856 deg | true heading 186.1 deg



Posterior and control metrics

MEAN POSITION ERROR

52.2 NM

MACH: POSTERIOR / TRUTH

0.805 / 0.753

GROUND SPEED: POSTERIOR / TRUTH

456 / 452 kt

TRUTH-IMPLIED BFO BIAS

171.6 Hz

POSTERIOR WITHIN 100 NM

87.8%

ALTITUDE: POSTERIOR / TRUTH

32344 / 28368 ft

LATENT BFO BIAS

168.2 Hz

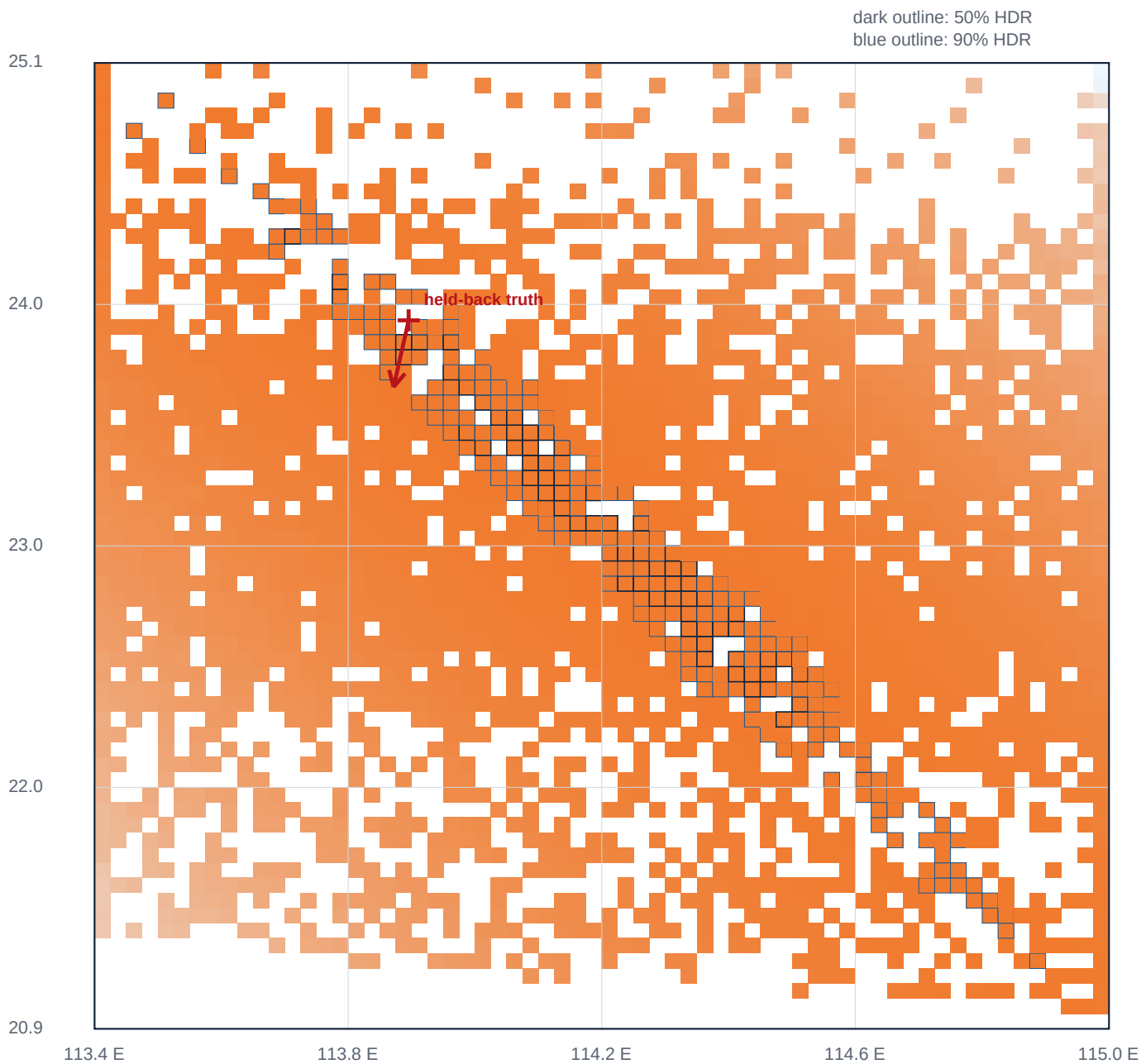
PARTICLE / ROOT ESS

45 / 44.8

Density combines equal-weight deterministic seed replicates. The red cross and arrow were loaded only during scoring.

mh371_0400 SATCOM arc

2014-03-07T04:04:57.916Z | truth 23.9533 deg, 113.8817 deg | true heading 192.7 deg



Posterior and control metrics

MEAN POSITION ERROR

59.6 NM

MACH: POSTERIOR / TRUTH

0.800 / 0.832

GROUND SPEED: POSTERIOR / TRUTH

461 / 455 kt

TRUTH-IMPLIED BFO BIAS

169.6 Hz

POSTERIOR WITHIN 100 NM

84.4%

ALTITUDE: POSTERIOR / TRUTH

32827 / 38100 ft

LATENT BFO BIAS

168.2 Hz

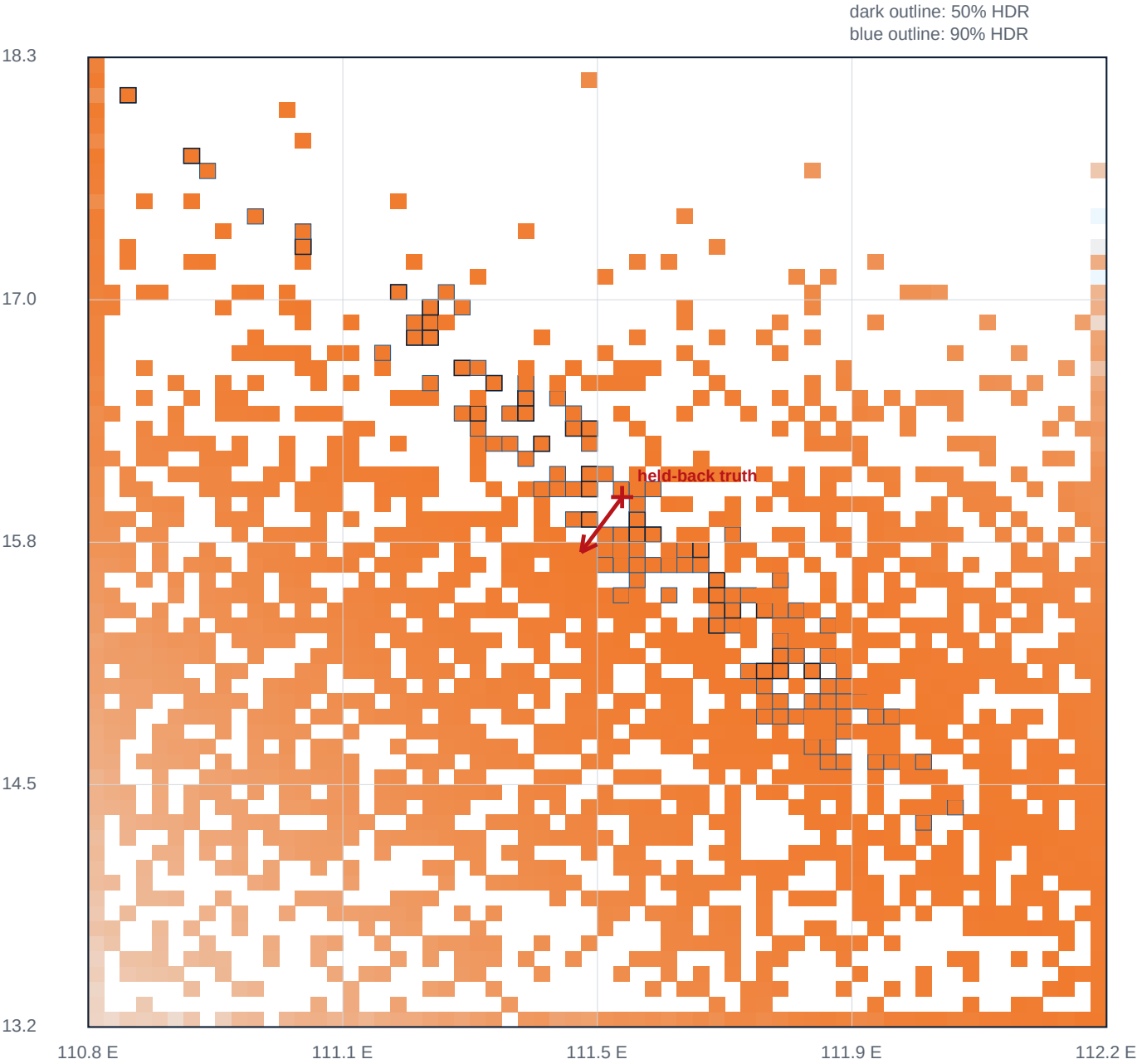
PARTICLE / ROOT ESS

235 / 37.7

Density combines equal-weight deterministic seed replicates. The red cross and arrow were loaded only during scoring.

mh371_0500 SATCOM arc

2014-03-07T05:11:37.411Z | truth 15.9956 deg, 111.5330 deg | true heading 216.7 deg



Posterior and control metrics

MEAN POSITION ERROR
3.9 NM

MACH: POSTERIOR / TRUTH
0.769 / 0.830

GROUND SPEED: POSTERIOR / TRUTH
440 / 462 kt

TRUTH-IMPLIED BFO BIAS
169.7 Hz

POSTERIOR WITHIN 100 NM
93.8%

ALTITUDE: POSTERIOR / TRUTH
35773 / 39998 ft

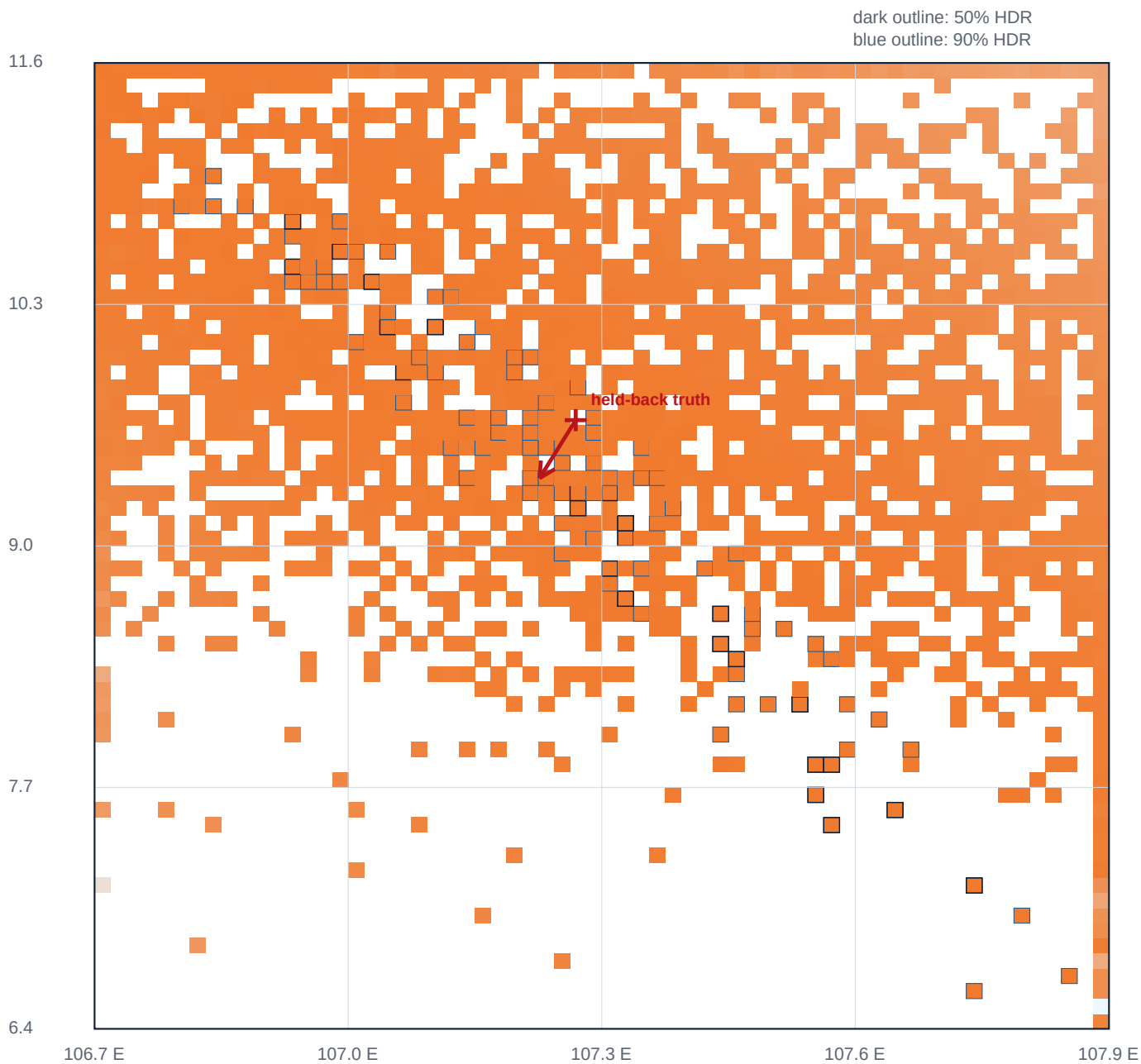
LATENT BFO BIAS
168.6 Hz

PARTICLE / ROOT ESS
60 / 25.2

Density combines equal-weight deterministic seed replicates. The red cross and arrow were loaded only during scoring.

mh371_0600 SATCOM arc

2014-03-07T06:09:44.409Z | truth 9.6835 deg, 107.2620 deg | true heading 212.0 deg



Posterior and control metrics

MEAN POSITION ERROR

33.6 NM

MACH: POSTERIOR / TRUTH

0.804 / 0.813

GROUND SPEED: POSTERIOR / TRUTH

484 / 474 kt

TRUTH-IMPLIED BFO BIAS

170.6 Hz

POSTERIOR WITHIN 100 NM

82.3%

ALTITUDE: POSTERIOR / TRUTH

33274 / 39997 ft

LATENT BFO BIAS

168.6 Hz

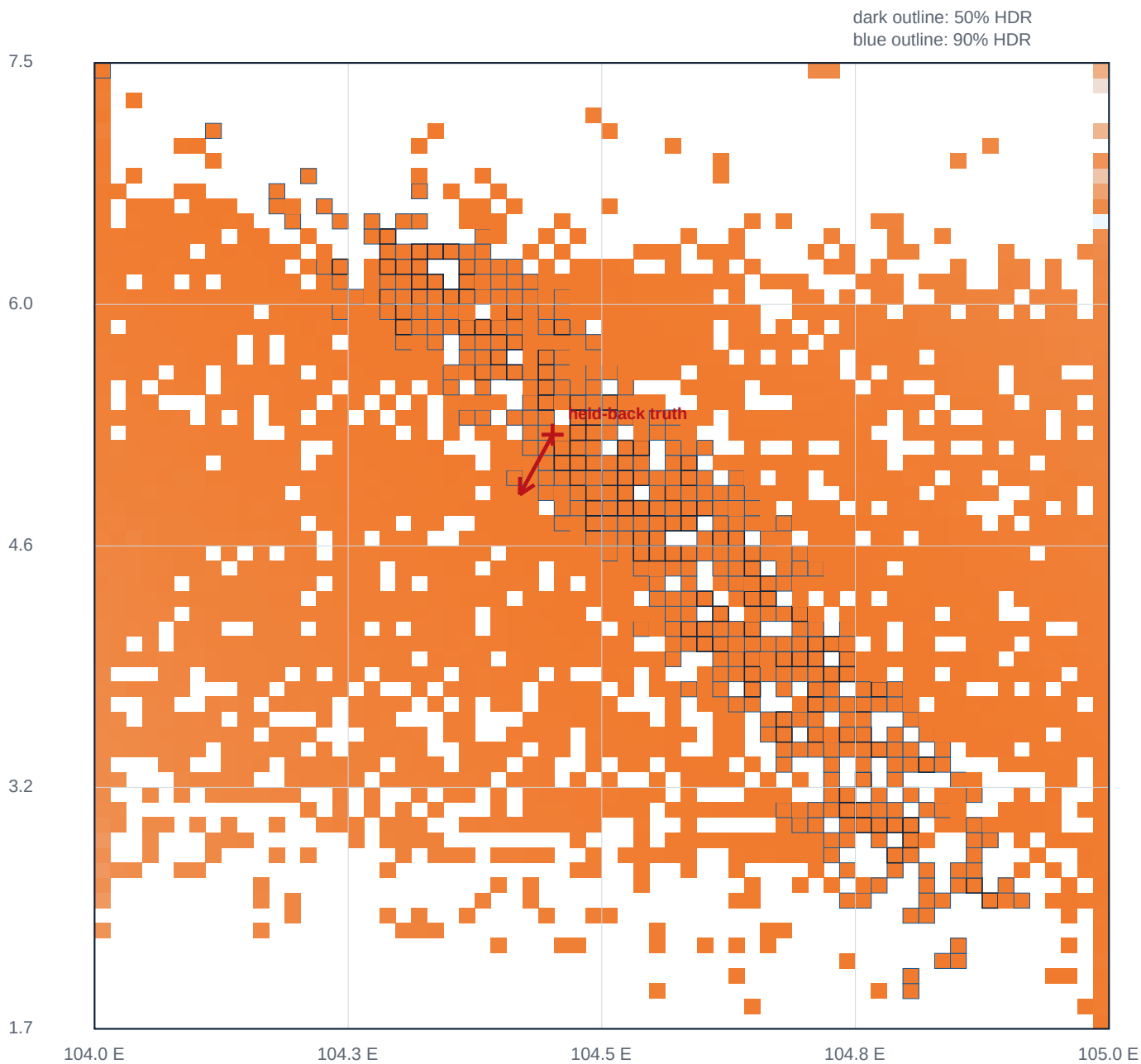
PARTICLE / ROOT ESS

42 / 20.6

Density combines equal-weight deterministic seed replicates. The red cross and arrow were loaded only during scoring.

mh371_0648 SATCOM arc

2014-03-07T06:48:33.907Z | truth 5.2504 deg, 104.4874 deg | true heading 208.8 deg



Posterior and control metrics

MEAN POSITION ERROR

37.0 NM

MACH: POSTERIOR / TRUTH

0.799 / 0.821

GROUND SPEED: POSTERIOR / TRUTH

486 / 483 kt

TRUTH-IMPLIED BFO BIAS

168.6 Hz

POSTERIOR WITHIN 100 NM

79.2%

ALTITUDE: POSTERIOR / TRUTH

33856 / 40000 ft

LATENT BFO BIAS

168.6 Hz

PARTICLE / ROOT ESS

381 / 18.6

Density combines equal-weight deterministic seed replicates. The red cross and arrow were loaded only during scoring.